

Scale-Adaptive Wasserstein–IoU for Tiny Object Detection

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Abstract

Detecting tiny objects is challenging because small localization errors can sharply reduce bounding-box overlap. Once predicted and target boxes become strictly disjoint, the plain IoU loss provides no centroid-gradient signal. We study Scale-Adaptive Wasserstein–IoU (H-WIoU), which blends an arm-specific geometric affinity with IoU through a rational gate without learned parameters, $\gamma(s) = s^2/(s^2 + \sigma_0^2)$. The frozen arms use either symmetric scale-normalized divergence or target-normalized Gaussian W_2 ; the YOLO path caps the latter’s exponent. For smaller targets, the objective places more weight on geometric affinity, while larger targets place more weight on IoU. For the two uncapped variants on compact, strictly disjoint box domains, we establish a positive centroid-gradient lower bound in exact real arithmetic and show that the negative centroid gradient is a descent direction for squared centroid error. We also study detached feature entropy (EH-WIoU) and apply the similarity to label assignment through HLA, H-TAL, and H-STAL.

Experiments on Scale Match TinyPerson and AI-TOD-v2 evaluate the formulation across Faster R-CNN, FCOS, Cascade R-CNN, RetinaNet, YOLOv8s, and NMS-free YOLO26s. On TinyPerson validation, Faster R-CNN dual placement increases AP_{50}^{tiny1} from 20.35% to 25.65%, while EH-WIoU reaches 26.46%; FCOS with H-WIoU and Gaussian centerness increases overall AP_{50} from 27.52% to 31.13% within its factorial protocol. Held-out TinyPerson test results include gains on the smallest-object subsets, while a further 51 extended models show results that vary by detector and metric. Across 71 canonical and extended test evaluations, these single-seed measurements do not establish a general ranking of losses. Validation and official-test scores are presented separately. The proposed terms are used only during training and introduce no additional inference-time operators. Code is available at the project repository.

1 Introduction

Detecting tiny objects is critical for real-world applications such as maritime search and rescue, drone surveillance, and aerial satellite monitoring [1, 2, 3]. In these settings, objects often occupy only a handful of pixels on the camera sensor. For example, a swimmer stranded in open water or a distant vehicle in satellite imagery may have a characteristic scale of $s = \sqrt{wh} < 8$ pixels.

At this microscopic scale, standard object detection methods face a fundamental geometric failure:

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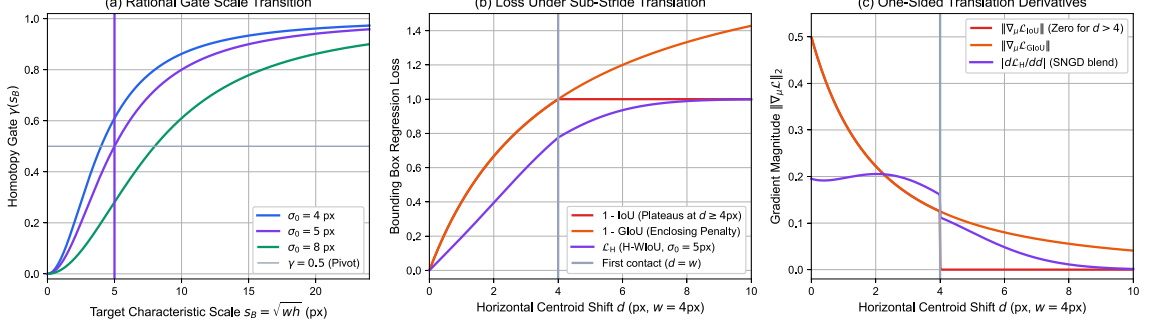


Figure 1: SNGD affinity geometry.

A shift of just 1 or 2 pixels can completely eliminate bounding-box overlap.

Consider two identical 4×4 -pixel bounding boxes. If one box is shifted by 2 pixels in both horizontal and vertical directions, the overlapping area drops from 16 pixels down to just 4 pixels—a 75% loss of overlap, resulting in $\text{IoU} = 4/(16 + 16 - 4) = 1/7 \approx 0.14$. If the shift exceeds 4 pixels, the boxes no longer overlap at all ($\text{IoU} = 0$).

Figure 1(a) shows how the rational scale gate changes the mixture between geometric affinity and IoU. Panel (b) translates equal 4×4 boxes horizontally: first contact occurs at $d = 4$ pixels, and strict separation begins for $d > 4$; the GIoU curve is not clipped. Panel (c) shows one-sided translation derivatives on smooth intervals, with unequal slopes for IoU and its mixture at contact. The plot includes overlap and contact, whereas the theorem’s uniform bound applies only on its compact, strictly separated domain.

Inside a strictly disjoint region, the IoU term is locally constant with respect to box coordinates, so its box-coordinate gradient is zero. Other training losses can still update the network; the missing signal here concerns the IoU regression term.

To address this problem, several geometry-extended losses have been proposed. Generalized IoU (GIoU) [4] adds a penalty based on the smallest enclosing box, and Distance-IoU (DIoU) [5] penalizes the distance between box centers. Their geometric penalties differ from the scale-conditioned affinity studied here; this paper compares the resulting detectors empirically rather than asserting a universal gradient direction for those losses.

Another approach is to model bounding boxes as 2D Gaussian distributions and measure their distance using the Normalized Wasserstein Distance (NWD) [6]. Its distance remains defined for separated boxes and provides a smooth center-distance term. A zero center displacement, however, has zero center gradient, and finite-precision exponentials can saturate at large distances. We blend geometric affinity with IoU so the weight assigned to each depends on target scale.

Our Insight: Scale-Adaptive Blending. We do not need to choose between IoU and Wasserstein distance. Instead, we can combine their strengths:

- For **tiny objects**, place more weight on a smooth, scale-normalized geometric affinity that remains positive at finite separation in exact arithmetic.
- For **larger objects**, rely primarily on standard IoU to ensure sharp, tight bounding boxes.

We realize this concept in **Scale-Adaptive Wasserstein-IoU (H-WIoU)** (Figure 1). Using a quadratic rational gate $\gamma(s) = s^2/(s^2 + \sigma_0^2)$, H-WIoU shifts mixture weight toward IoU as target scale grows without adding learned gate parameters; the pivot σ_0 is selected by protocol. The frozen implementations use either a symmetric scale-normalized divergence or target-normalized Gaussian Wasserstein affinity, as specified below. We also study an entropy-guided extension (**EH-WIoU**) and apply scale-adaptive similarity to positive label assignment (**HLA**, **H-TAL**, and **H-STAL**).

Key Contributions:

1. **Scale-Adaptive Formulation.** We study H-WIoU as a family that combines one of two specified geometric affinities with IoU through a rational gate controlled by target scale, with no learned gate parameters.
2. **Analytical Properties.** On compact, strictly disjoint box domains, we establish a uniform positive centroid-gradient lower bound in exact real arithmetic and show that the negative centroid gradient is a descent direction for squared centroid error.
3. **Detector Integration and Factorial Analysis.** We integrate the similarity into regression and assignment across Faster R-CNN, FCOS, Cascade R-CNN, RetinaNet, YOLOv8s, and YOLO26s, and use factorial experiments to separate loss, centerness, sampling, and placement effects.
4. **Held-Out Evaluation.** We evaluate selected configurations on the official TinyPerson test split and report detector-dependent gains and trade-offs across fine-grained object-scale subsets.

2 Related Work

2.1 Bounding-Box Regression Losses

Early object detectors used Smooth-L1 loss to penalize coordinate offsets independently [7]. Because independent coordinate regression does not optimize overall box overlap, IoU loss was introduced to directly align training with evaluation metrics. To handle disjoint boxes where IoU is zero, GIoU [4] incorporates the area of the smallest enclosing box, DIoU [5] adds a normalized center-distance penalty, and CIoU [5] / EIoU [8] add aspect-ratio and dimension penalties. Other variants, such as α -IoU [9], Wise-IoU [10], SIoU [11], and Inner-IoU [12], reshape gradient weights or use auxiliary scaling factors.

In parallel, Wasserstein-based losses model bounding boxes as 2D Gaussians. Normalized Wasserstein Distance (NWD) [6, 3] computes a transport-based similarity that remains defined when boxes do not overlap. Subsequent works, including GWD [15], KLD [16], IGWD [17], and GCD [18], extended Gaussian modeling to oriented boxes and aerial imagery. H-WIoU blends an identified geometric affinity with IoU as a continuous function of object scale.

2.2 Label Assignment for Tiny Objects

Assigning positive training anchors is notoriously difficult for tiny objects. Standard Max-IoU matching requires candidates to exceed fixed overlap thresholds (e.g., 0.5 or 0.7), which microscopic objects rarely achieve. Adaptive strategies such as ATSS [22], PAA [23], and OTA [24]

compute statistical or optimal-transport assignments. For tiny objects specifically, RFLA [25] uses Gaussian receptive-field points, and SimD [26] combines centroid distance with shape similarity. Our Homotopy Label Assignment (HLA) incorporates scale-adaptive similarity directly into the anchor-matching stage.

2.3 Real-Time and NMS-Free Detectors

Modern one-stage detectors like YOLOv8 [29] and YOLOv9 [28] use the Task-Aligned Assigner (TAL) [27], which ranks candidates by $t = s^\alpha \cdot \text{IoU}^\beta$. End-to-end NMS-free models, such as YOLO26s [34] (built upon dual-label assignment [33]), eliminate non-maximum suppression by combining one-to-one and one-to-many matching branches. In this paper, we extend our scale-adaptive similarity into both YOLOv8s (H-TAL) and YOLO26s (H-STAL).

3 Methodology

3.1 Representing Boxes as 2D Gaussians

Let a predicted box be $A = (\mu_a, w_a, h_a)$ and its target be $B = (\mu_b, w_b, h_b)$, where $\mu = (x, y)^T$ is the box center, and $w, h > 0$ are the width and height. Following NWD [6], we can associate each box with a 2D Gaussian distribution $\mathcal{N}(\mu, \Sigma)$, where the covariance matrix is defined by:

$$\Sigma = \text{diag} \left(\frac{w^2}{4}, \frac{h^2}{4} \right). \quad (1)$$

This converts a crisp, hard-edged rectangle into a smooth, continuous bell curve. The 2-Wasserstein distance between two such Gaussian distributions is:

$$\mathcal{W}_2^2(\mathcal{N}_a, \mathcal{N}_b) = \|\mu_a - \mu_b\|_2^2 + \frac{(w_a - w_b)^2 + (h_a - h_b)^2}{4}. \quad (2)$$

Notice that this distance has physical units of pixels squared. A 2-pixel error on a 4-pixel object has the same numerical penalty as a 2-pixel error on a 100-pixel object, even though a 2-pixel shift is massive for the tiny object (50% of its size) and negligible for the large object (2%). To fix this, we must normalize the distance by scale.

3.2 Geometric Affinities Used by the Frozen Arms

One implementation defines the **Scale-Normalized Geometric Divergence (SNGD)** by normalizing center displacements with both boxes' dimensions and measuring shape by log ratios:

$$\mathcal{D}_{\text{SN}}^2(A, B) = \frac{2(x_a - x_b)^2}{w_a^2 + w_b^2} + \frac{2(y_a - y_b)^2}{h_a^2 + h_b^2} + \ln^2 \left(\frac{w_a}{w_b} \right) + \ln^2 \left(\frac{h_a}{h_b} \right). \quad (3)$$

This divergence has several appealing properties:

- **Scale Invariance:** If we scale both boxes by a constant factor $\lambda > 0$ (e.g., zooming in on an image), $\mathcal{D}_{\text{SN}}^2$ remains completely unchanged.
- **Non-negativity:** $\mathcal{D}_{\text{SN}}^2(A, B) \geq 0$, and equals 0 if and only if $A = B$.

- **Symmetry:** $\mathcal{D}_{\text{SN}}^2(A, B) = \mathcal{D}_{\text{SN}}^2(B, A)$.

Its affinity is bounded in $(0, 1]$ via the exponential map:

$$T_{\text{SN}}(A, B) = \exp(-\mathcal{D}_{\text{SN}}^2(A, B)). \quad (4)$$

The other implementation normalizes the squared Gaussian Wasserstein distance in Eq. (2) by the *target* area. With $s_B^2 = w_b h_b$ and numerical stabilizer $\varepsilon > 0$, it computes

$$\mathcal{D}_{\text{TW}}^2(A, B) = \frac{\mathcal{W}_2^2(\mathcal{N}_a, \mathcal{N}_b)}{2s_B^2 + \varepsilon}, \quad T_{\text{TW}}(A, B) = \exp(-\mathcal{D}_{\text{TW}}^2(A, B)). \quad (5)$$

Unlike SNGD, this target-normalized affinity uses squared width and height differences and is asymmetric in candidate and target. Both affinities equal one for identical boxes and remain positive at finite displacement in exact arithmetic. YOLO’s implementation caps the exponent at 30 for numerical stability; that cap changes the gradient at extreme separations. We write T_q for the exact affinity $q \in \{\text{SN}, \text{TW}\}$ selected by the frozen arm and identify the YOLO capped variant separately.

3.3 The Rational Scale Gate

To blend the selected affinity $T_q(A, B)$ with standard IoU(A, B), we define a scale-dependent gating function. Let $s_B = \sqrt{w_b h_b}$ denote the scale of the target box, and let $\sigma_0 > 0$ be a reference scale pivot (in pixels). We define the rational gate $\gamma(s_B)$ as:

$$\gamma(s_B) = \frac{s_B^2}{s_B^2 + \sigma_0^2}, \quad 1 - \gamma(s_B) = \frac{\sigma_0^2}{s_B^2 + \sigma_0^2}. \quad (6)$$

This gate behaves in an intuitive and desirable way:

1. For **microscopic objects** ($s_B \rightarrow 0$), $\gamma(s_B) \rightarrow 0$ and $1 - \gamma(s_B) \rightarrow 1$. The supervision relies entirely on the selected geometric affinity $T_q(A, B)$.
2. For **large objects** ($s_B \gg \sigma_0$), $\gamma(s_B) \rightarrow 1$ and $1 - \gamma(s_B) \rightarrow 0$. The supervision transitions completely to standard IoU(A, B).
3. At the **pivot scale** ($s_B = \sigma_0$), $\gamma(\sigma_0) = 0.5$, providing an equal 50/50 blend.

For either identified affinity, the **H-WIoU similarity** and its corresponding regression loss are:

$$\begin{aligned} \mathcal{S}_{\text{H},q}(A, B) &= \gamma(s_B) \cdot \text{IoU}(A, B) + (1 - \gamma(s_B)) \cdot T_q(A, B), \\ \mathcal{L}_{\text{H},q}(A, B) &= 1 - \mathcal{S}_{\text{H},q}(A, B). \end{aligned} \quad (7)$$

Both $\mathcal{S}_{\text{H},q}$ and $\mathcal{L}_{\text{H},q}$ are bounded within $[0, 1]$. On strictly separated boxes with nonzero center displacement, the uncapped affinities supply a nonzero center-gradient component in exact arithmetic even though the IoU component is locally constant.

3.4 Optional Feature Entropy Extension (EH-WIoU)

In complex backgrounds (such as sea clutter or aerial shadows), we can optionally modulate the gate using the channel entropy of local deep features. For a feature vector at spatial location q with $C_f > 1$ channels, we compute its normalized channel entropy $H_q \in [0, 1]$:

$$p_{q,c} = \frac{e^{f_{q,c}}}{\sum_{j=1}^{C_f} e^{f_{q,j}}}, \quad H_q = -\frac{\sum_{c=1}^{C_f} p_{q,c} \ln p_{q,c}}{\ln C_f}. \quad (8)$$

We then modulate the scale gate using a coupling factor $\beta \geq 0$:

$$\gamma_\beta(s_B, H) = \frac{s_B^2(1 + \beta H)}{s_B^2(1 + \beta H) + \sigma_0^2}. \quad (9)$$

When $\beta = 0$, this recovers the scale-only gate for the *same selected affinity* q . During backpropagation, detaching entropy stops gradients through the entropy-prior computation; the regression or assignment loss can still update shared detector parameters through other paths.

3.5 Theoretical Guarantees

Theorem 1 (Disjoint-box centroid gradient) *Fix a target box B with $w_b, h_b > 0$ and $\sigma_0 > 0$. Define*

$$\mathcal{Z}_B = \{A : \text{IoU}(A, B) = 0\}.$$

For $0 < \delta \leq \kappa < \infty$ and $0 < r_{\min} \leq r_{\max} < \infty$, let K be a nonempty compact subset of $\text{int}(\mathcal{Z}_B)$ such that every $A \in K$ satisfies

$$\delta \leq \frac{\|\mu_a - \mu_b\|_2}{s_B} \leq \kappa, \quad \frac{w_a}{w_b} \in [r_{\min}, r_{\max}], \quad \frac{h_a}{h_b} \in [r_{\min}, r_{\max}]. \quad (10)$$

For either uncapped, scale-only affinity $q \in \{\text{SN}, \text{TW}\}$, in exact real arithmetic there exists a constant $c_q > 0$, depending on B , the stated bounds, σ_0 , and q , such that

$$\|\nabla_{\mu_a} \mathcal{L}_{H,q}(A, B)\|_2 \geq c_q \quad (11)$$

uniformly for all $A \in K$. This statement does not cover YOLO’s capped exponent or finite-precision saturation.

Intuition of Proof. For $A \in K \subset \text{int}(\mathcal{Z}_B)$, the boxes are strictly disjoint, so the IoU term is locally constant. Differentiating the two uncapped affinities with respect to the predicted center gives

$$g_{\text{SN}} = 4(1 - \gamma)T_{\text{SN}} \begin{bmatrix} (x_a - x_b)/(w_a^2 + w_b^2) \\ (y_a - y_b)/(h_a^2 + h_b^2) \end{bmatrix}, \quad (12)$$

$$g_{\text{TW}} = \frac{2(1 - \gamma)T_{\text{TW}}}{2s_B^2 + \varepsilon}(\mu_a - \mu_b).$$

Because $1 - \gamma > 0$, each uncapped $T_q > 0$, and $\|\mu_a - \mu_b\|_2 \geq \delta s_B > 0$ with bounded positive dimensions, each continuous gradient norm has a positive minimum on compact K . Figure 1(c) illustrates one translation slice of the SNGD variant; it does not plot the uniform constant c_q . $\square$

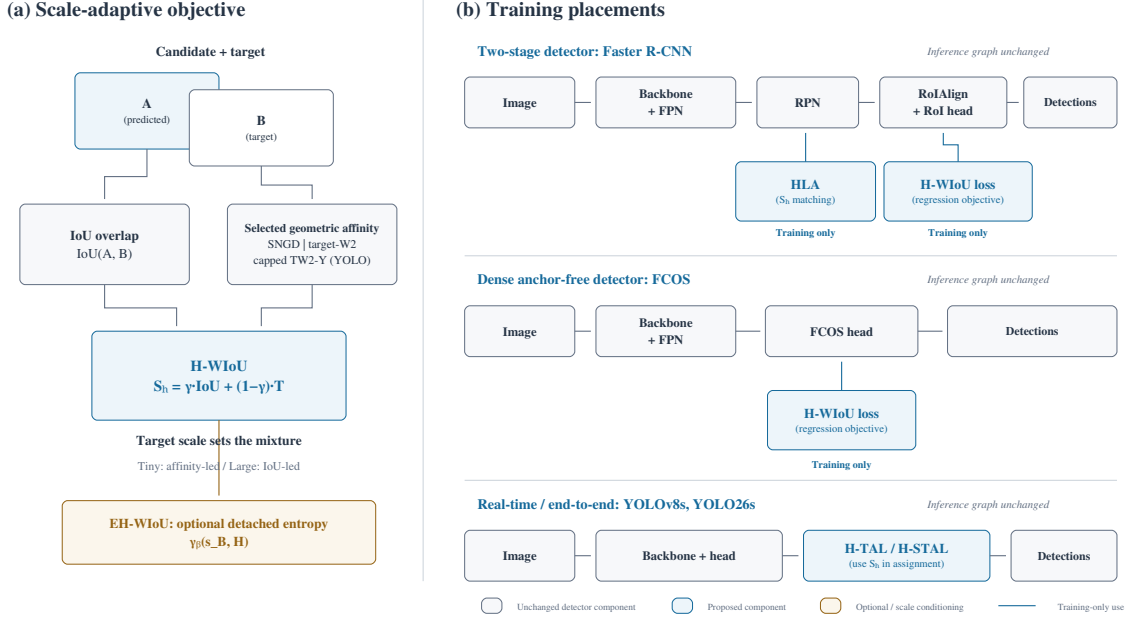


Figure 2: H-WIoU formulations and detector integration.

Proposition 1 (Corrective Gradient Direction) *For either uncapped scale-only affinity $q \in \{\text{SN}, \text{TW}\}$, any strictly disjoint differentiable box A with $\Delta = \mu_a - \mu_b \neq 0$ has a centroid gradient $g_q = \nabla_{\mu_a} \mathcal{L}_{H,q}$ satisfying*

$$\langle g_q, \Delta \rangle > 0. \quad (13)$$

Each expression in Eq. (12) is a positive coefficient times the displacement, with positive coordinate weights for SNGD. Thus $-g_q$ is a strict descent direction for squared centroid error. A centroid-only update

$$\mu_a^+ = \mu_a - \alpha g_q \quad (14)$$

reduces the squared centroid error for sufficiently small $\alpha > 0$. This statement concerns the box-center variables and does not guarantee improvement under an arbitrary network-parameter update.

3.6 Detector Integration

As shown in Figure 2, the frozen arms use different affinities and training placements. FCOS, the Faster R-CNN placement and beta factorials, Cascade R-CNN, and RetinaNet use the target-normalized W_2 path. YOLO uses its capped target-normalized path. The generic Faster R-CNN metric registry can instead select SNGD under `h_wiou`; each reported arm must be identified from its frozen command and config. These training changes leave the compared inference architectures unchanged:

- **RPN Label Assignment (HLA):** In Faster R-CNN, the identified $\mathcal{S}_{H,q}$ replaces the standard IoU score used during anchor matching. This can change candidate ranking and threshold crossings while retaining the detector’s remaining matching rules.

- **RoI / Dense Box Regression:** The identified $\mathcal{L}_{H,q}$ replaces the configured baseline box loss in Faster R-CNN, FCOS, Cascade R-CNN, or RetinaNet, with placement specified per arm.
- **Task-Aligned Assignment (H-TAL & H-STAL):** In one-stage YOLOv8s and end-to-end NMS-free YOLO26s, the alignment score $t = s^\alpha \cdot \text{IoU}^\beta$ is replaced with:

$$t_H = s^\alpha \cdot [\mathcal{S}_{H,q}(A, B)]^\beta. \quad (15)$$

For finite distances below YOLO’s exponent cap, this avoids forcing the spatial component to zero solely because IoU is zero.

4 Experimental Setup

4.1 Datasets and Evaluation Metrics

- **Scale Match TinyPerson [1]:** A benchmark for tiny person detection in maritime and seaside environments. We use the audited Scale Match surface with 628 training and 118 validation images, tiled into 512×512 patches with 64-pixel overlap (1,684 validation tiles). We evaluate overall AP at IoU thresholds of 0.25, 0.50, and 0.75, along with fine-grained scale subsets: $\text{AP}_{50}^{\text{tiny1}}$ ($s \leq 8$ px) and reasonable-scale objects ($\text{AP}_{50}^{\text{reas.}}$).
- **AI-TOD-v2 [3]:** The dataset authors report 28,036 aerial images across the released splits. Our frozen training contract contains 11,214 training annotation records (11,204 nonempty workload images) and 2,804 allowed validation images; the held-out final split is evaluated only in the separately audited official-test results. We report AP (averaged across $\text{IoU} \in [0.50 : 0.05 : 0.95]$), AP_{50} , AP_{75} , very-tiny and tiny-object AP (AP_{vt} and AP_t), small-object AP_s , and recall AR_{1500} .

All metrics in this paper are reported in **percentages (%)** with two decimal places, and IoU thresholds are ordered in ascending sequence ($\text{AP}_{25} \rightarrow \text{AP}_{50} \rightarrow \text{AP}_{75}$).

4.2 Implementation and Training Protocols

All experiments use PyTorch and Seed 42. Training runs were executed on Kaggle GPUs; an NVIDIA RTX 5070 Ti was used for local checkpoint replay and latency measurements. Two-stage and dense anchor-free models use ResNet-50-FPN backbones [35, 20]. Model parameters are: Faster R-CNN (41.3M), FCOS (32.1M), Cascade R-CNN (41.9M), RetinaNet (32.2M), YOLOv8s (11.1M), and YOLO26s (9.9M). The TinyPerson Faster R-CNN, FCOS, Cascade R-CNN, and RetinaNet comparisons use 20-epoch training budgets; family-specific optimizer settings are recorded in their frozen protocols. AI-TOD-v2 FCOS loss comparisons use 10 epochs, whereas the Faster R-CNN label-assignment comparison uses 12 epochs. YOLOv8s and YOLO26s request a maximum of 100 epochs with early-stopping patience 25; some runs end earlier.

Table 1: TinyPerson validation across detector families.

Detector / Configuration	Parameters	AP ₂₅ ^{all}	AP ₅₀ ^{all}	AP ₇₅ ^{all}	AP ₅₀ ^{tiny1}	AP ₅₀ ^{reas.}
Faster R-CNN: Baseline	41.3M	59.30	42.22	5.92	20.35	54.59
HLA + RoI H-WIoU (Dual)	41.3M	59.48	42.22	6.05	25.65	61.75
EH-WIoU ($\beta = 1$)	41.3M	59.42	42.97	6.15	26.46	59.78
FCOS: GIoU Baseline	32.1M	48.26	27.52	3.05	5.58	43.45
H-WIoU + Gaussian centerness	32.1M	54.20	31.13	3.06	8.10	51.26
H-WIoU + Gaussian centerness + TOD sampling	32.1M	55.75	30.10	3.06	11.92	41.47
Cascade R-CNN: Baseline	41.9M	61.86	46.08	7.00	26.70	58.51
EH-WIoU ($\sigma_0 = 5$, $\beta = 0.5$)	41.9M	60.97	44.81	6.34	—	60.56
RetinaNet: Baseline	32.2M	51.04	34.58	4.12	7.82	39.80
H-WIoU ($\sigma_0 = 5$)	32.2M	51.86	35.12	4.25	8.94	42.12

5 Results and Analysis

5.1 TinyPerson Multi-Paradigm Evaluation

Table 1 reports performance across four detector paradigms on Scale Match TinyPerson. Boldface marks peak displayed values within each detector family.

Key observations from Table 1:

- **Faster R-CNN:** Dual placement (HLA in RPN + H-WIoU in RoI) improves microscopic object detection, raising AP₅₀^{tiny1} from 20.35% to 25.65% (+5.30 percentage points, a 26% relative gain) and reasonable-scale AP from 54.59% to 61.75% (+7.16 pp). The entropy-guided variant (EH-WIoU, $\beta = 1$) achieves the highest overall AP₅₀ of 42.97% and peak AP₅₀^{tiny1} of 26.46%.
- **FCOS:** The standard GIoU baseline reaches 27.52% AP₅₀. H-WIoU with Gaussian centerness reaches 31.13% AP₅₀ (+3.61 pp) and 51.26% reasonable AP (+7.81 pp). Adding TOD-adaptive sampling gives 11.92% AP₅₀^{tiny1}, compared with 5.58% for the baseline.
- **Cascade R-CNN & RetinaNet:** The Cascade method’s clean EMA replay gives 60.97% AP₂₅, 44.81% overall AP₅₀, and 60.56% reasonable-scale AP₅₀, versus 61.86%, 46.08%, and 58.51% for its baseline. Its tiny1 AP₅₀ remains withheld because two replay records differ beyond tolerance. RetinaNet’s tiny1 AP₅₀ rises from 7.82% to 8.94% in its tested comparison.

5.2 Sensitivity to the H-WIoU Homotopy Form

We compared five alternative homotopy forms with the accepted rational-form control on the same Seed-42 TinyPerson validation surface. All models use H-WIoU with $\sigma_0 = 5$ pixels, standard centerness and sampling, and a 20-epoch schedule. Table 2 reports each run’s best validation checkpoint; the rational control was independently replayed in the factorial study. Boldface marks the highest observed value for each metric among the six forms. The sigmoid form has the highest observed AP, AP₅₀, and tiny1 AP₅₀. The rational form is therefore not the top-performing choice in this single-seed comparison; these results describe the tested checkpoints and do not establish statistical significance.

Table 2: Homotopy-form sensitivity on TinyPerson validation.

Homotopy form	Best epoch	AP	AP ₅₀	AP ₅₀ ^{tiny1}
Rational	6	25.74	26.23	5.66
Exponential	6	25.89	26.67	6.50
Sigmoid	10	26.90	27.93	8.18
Static ($\gamma = 0.5$)	8	25.13	25.52	6.86
Pure W ₂ ($\gamma = 0$)	10	22.12	22.27	5.35
Pure IoU ($\gamma = 1$)	9	26.48	27.06	8.04

Table 3: Bounding-box loss comparison on TinyPerson validation.

Category	Loss Objective	Venue	Scale Match TinyPerson (20 Epochs)			
			AP ₅₀ ^{all}	AP ₇₅ ^{all}	AP ₅₀ ^{tiny1}	AP ₅₀ ^{reas.}
Standard	GIoU	CVPR'19	27.52	3.05	5.58	43.45
Geometric	CIoU	AAAI'20	23.93	2.41	7.91	35.70
	DIoU	AAAI'20	25.46	2.81	8.58	39.58
	EIoU	Neurocomp.'22	24.23	1.89	7.66	42.34
	WIoU v1	arXiv'23	24.42	2.38	9.89	35.66
Distributional	NWD	arXiv'21	22.83	1.78	7.76	41.08
	IGWD (Best AP ₅₀ , 6)	TMM'26	26.43	2.99	4.85	43.59
	SimD	IROS'24	23.51	2.19	7.32	39.15
	GCD (Best AP ₅₀ , 15)	GRSL'25	24.87	1.87	7.93	37.37
Extended	α -IoU	NeurIPS'21	26.54	2.79	9.41	44.82
	SIoU	arXiv'22	23.69	2.00	8.19	39.77
	Inner-IoU (Best AP ₅₀ , 6)	arXiv'23	27.07	3.12	5.67	46.93
Proposed	H-WIoU (Loss Only)	Ours	26.23	2.90	5.66	40.48
	H-WIoU + Gaussian centerness	Ours	31.13	3.06	8.10	51.26
	H-WIoU + Gaussian centerness + TOD sampling	Ours	30.10	3.06	11.92	41.47

5.3 Comparison with Published Bounding-Box Objectives

Table 3 summarizes Seed-42 TinyPerson validation comparisons. TinyPerson factorial rows use best-validation checkpoints. External TinyPerson objectives use terminal epoch-20 metrics except IGWD, GCD, and Inner-IoU, whose frozen best-AP₅₀ checkpoints are identified in the row labels and pass selector-bound replay. Prediction-schema recovery removes only out-of-contract class index 0 from separate derived files; evaluator metrics are unchanged. Because the selectors differ, these results are descriptive rather than a cross-method ranking; boldface marks column maxima only. Rows that fail scoped replay or schema gates are withheld, and Gaussian-centerness rows are system-level comparisons.

The selected TinyPerson factorial checkpoints report AP₅₀ values of 26.23% for H-WIoU loss only and 31.13% for H-WIoU with Gaussian centerness. IGWD, GCD, and Inner-IoU use predeclared best-AP₅₀^{all} checkpoints (epochs 6, 15, and 6); all pass strict reload and full 1,684-tile replay at the unchanged 0.0005 tolerance, while their terminal selectors remain held. Other external rows use terminal epoch-20 metrics and separate manifest-bound best-checkpoint predictions for schema checks; the three rows labeled Best AP₅₀ use the same selector for both. These mixed-selector results are descriptive only. AI-TOD-v2 loss rows remain audit-only because no complete protocol-matched comparison passed identity, schema, and replay gates. Gaussian-centerness configurations change the training setup and do not isolate the loss effect.

5.4 Comparison of Label-Assignment Strategies

Table 4 reports Seed-42 TinyPerson Faster R-CNN assignment diagnostics and marks the checkpoint selector used for each row. Standard IoU and HLA use placement-factorial best checkpoints; ATSS and SimD use terminal epoch-20 checkpoints; RFLA uses its predeclared best-AP50 checkpoint at epoch 6. Every displayed row passes full-validation replay for its listed selector. ATSS and SimD schema checks use manifest-bound best-checkpoint predictions separate from their displayed terminal metric selectors.

NWD-assignment remains withheld because neither terminal nor best-checkpoint replay passes. AI-TOD-v2 values remain audit-only because no assignment row has a complete accepted identity and replay chain. These scores are descriptive; selector differences prevent a cross-method ranking, and boldface only marks the highest displayed value in each metric column.

Table 4: Assignment strategies in TinyPerson Faster R-CNN.

Assignment Strategy	Selector	Venue	TinyPerson (20 Epochs)			
			AP_{50}^{all}	AP_{75}^{all}	AP_{50}^{tiny1}	$AP_{50}^{\text{reas.}}$
Standard IoU Max-Match	Best	NeurIPS’15	42.22	5.92	20.35	54.59
ATSS	Terminal (20)	CVPR’20	35.58	3.96	16.13	49.58
RFLA	Best AP_{50} (6)	ECCV’22	39.45	4.99	15.82	54.81
NWD-assign	Withheld	arXiv’21	—	—	—	—
SimD-assign	Terminal (20)	IROS’24	33.34	3.14	18.03	48.94
HLA (RPN Only)	Placement best	Ours	42.48	6.12	22.59	54.52
Dual HLA (RPN + RoI)	Placement best	Ours	42.22	6.05	25.65	61.75

RFLA’s terminal epoch-20 replay misses the tiny1 AP_{50} tolerance by 0.000659224; Table 4 therefore reports its predeclared, replay-passing best AP_{50} checkpoint at epoch 6. AI-TOD-v2 assignment values remain audit-only because their source/config and selector-bound replay chains are incomplete.

5.5 Real-Time YOLOv8s and NMS-Free YOLO26s Results

Table 5 reports native Ultralytics validation diagnostics from the last logged training epoch of each YOLO run. These four metrics per dataset share the same native evaluator; they are not the independent benchmark replay scores of the saved `best.pt` checkpoints. The requested training maximum is 100 epochs with patience 25, but early stopping shortened several TinyPerson runs. The models are YOLOv8s (11.1M parameters) and NMS-free YOLO26s (9.9M parameters); bold-face marks the highest native metric within each detector family. We interpret differences within each detector family as descriptive native-validation results.

Table 5: YOLO validation metrics at the last logged epoch.

Detector	Formulation	AI-TOD-v2 (up to 100 epochs)				TinyPerson (up to 100 epochs)			
		Precision	Recall	AP ₅₀	AP	Precision	Recall	AP ₅₀	AP
YOLOv8s	CIoU Baseline	63.55	51.80	54.08	24.18	58.26	41.42	44.17	15.21
	IGWD	61.91	52.33	54.57	24.09	58.08	40.64	43.77	14.75
	H-WIoU	64.61	51.48	53.95	23.78	58.65	38.16	40.75	13.67
	H-TAL (Dual)	58.47	54.32	53.89	22.96	56.15	35.79	39.13	12.83
YOLO26s (NMS-Free)	CIoU Baseline	56.93	53.33	53.44	23.71	56.29	45.74	47.45	16.03
	IGWD	60.99	54.20	53.98	23.27	54.61	41.79	44.68	14.90
	H-WIoU	59.09	51.25	52.63	22.45	56.13	39.61	42.46	14.56
	H-STAL (Dual)	72.01	50.04	53.32	22.70	52.35	39.77	41.53	13.73

In the native AI-TOD-v2 YOLO26s logs, H-STAL has precision 72.01% versus 56.93% for CIoU (+15.08 pp), with lower recall (50.04% versus 53.33%) and similar native AP₅₀ (53.32% versus 53.44%). The corresponding TinyPerson H-STAL run has lower native AP than its CIoU baseline. These diagnostic contrasts do not establish a common-evaluator benchmark ranking; independent replay of saved best checkpoints is tracked separately.

5.6 Official TinyPerson Benchmark Test Evaluation

We evaluated models on the official held-out TinyPerson test set (786 images, 18,508 ground-truth annotations) using the official PointTinyBenchmark evaluator with ignore-uncertain annotations enabled. Sliding-window inference uses 512×512 tiles with 64-pixel overlap (23,319 tiles per model). In Table 6, boldface marks peak scores within each detector family.

The held-out TinyPerson test set shows the clearest gains on the smallest-object subsets for Faster R-CNN and FCOS. Faster R-CNN EH-WIoU Dual increases AP₅₀^{tiny1} from 26.68% to 30.92% (+4.24 pp), while the FCOS H-WIoU bundle increases it from 9.95% to 15.71% (+5.76 pp). FCOS H-WIoU with Gaussian centerness also increases overall AP₅₀ from 40.07% to 43.18%. Cascade R-CNN shows nearly unchanged tiny1 AP but lower aggregate AP₅₀/AP₇₅, illustrating that the benefits are not uniform across detector families.

5.7 AI-TOD-v2 Official Test Results

The canonical AI-TOD-v2 test results cover all 14,018 images and are separate from the validation results reported earlier. Boldface marks the highest score within each detector family.

Table 6: Canonical TinyPerson official-test results.

Detector	Formulation	AP ₂₅ ^{all}	AP ₅₀ ^{all}	AP ₇₅ ^{all}	AP ₅₀ ^{tiny}	AP ₅₀ ^{tiny1}	AP ₅₀ ^{tiny2}	AP ₅₀ ^{tiny3}	AP ₅₀ ^{small}	AP ₅₀ ^{reas.}
Faster R-CNN	Standard Baseline	70.17	52.33	9.83	47.24	26.68	50.78	61.12	69.47	70.33
	EH-WIoU Dual	71.79	53.23	9.92	49.53	30.92	53.39	61.59	68.99	68.01
	Unit Entropy ($\beta = 1$)	71.84	52.98	9.69	49.34	30.92	53.38	62.02	68.62	67.52
FCOS	GIoU Baseline	63.62	40.07	5.62	31.95	9.95	30.71	47.91	61.10	62.35
	H-WIoU + Gaussian centerness	68.31	43.18	6.35	36.79	11.68	38.09	53.05	61.62	62.31
	H-WIoU + Gaussian centerness + TOD sampling	69.56	43.16	6.02	37.59	15.71	38.46	52.59	60.76	60.00
Cascade R-CNN	Baseline	72.61	55.33	11.03	52.41	32.18	56.79	65.10	70.25	70.17
	Cascade EH-WIoU	72.34	54.87	10.40	51.15	32.20	54.57	64.25	70.45	70.03
YOLOv8s	H-TAL (Dual)	71.87	51.69	7.82	49.03	33.86	55.29	61.36	64.48	59.18
YOLO26s	H-STAL (NMS-Free)	72.54	52.42	8.38	48.91	34.98	55.72	60.17	64.84	61.34

Table 7: Canonical AI-TOD-v2 official-test results.

Detector	Formulation	AP	AP ₅₀	AP ₇₅	AP _{vt}	AP _t	AP _s	AR ₁₅₀₀
FCOS	GIoU baseline	16.65	39.81	11.01	4.05	15.04	21.48	28.27
FCOS	H-WIoU loss-only	17.16	41.43	11.30	3.26	16.02	21.82	28.45
FCOS	H-WIoU bundle	17.61	43.09	11.41	5.26	16.42	22.04	28.87
Faster R-CNN	Standard baseline	17.90	43.79	11.67	5.56	17.20	23.31	27.98
Faster R-CNN	HLA RPN	21.26	48.58	15.04	8.91	21.32	24.52	31.33
Faster R-CNN	H-WIoU Dual (entropy prior disabled)	17.98	43.90	11.25	6.17	17.60	22.32	28.32
YOLOv8s	CIoU baseline	21.71	49.86	15.20	7.08	19.86	28.41	32.00
YOLOv8s	H-TAL	20.79	49.83	13.61	6.21	18.67	28.65	31.92
YOLO26s	CIoU baseline	21.15	48.78	14.70	8.31	20.03	27.60	31.11
YOLO26s	H-STAL (NMS-free)	21.14	50.36	13.93	7.63	20.04	27.50	31.35

5.8 Additional Official-Test Results

The extended evaluation adds 51 models to the 20 canonical models: 22 additional AI-TOD-v2 models evaluated on 14,018 images and 29 additional TinyPerson models evaluated on 786 images. Each row is one Seed-42 checkpoint evaluated on its official test split. These tables describe the observed models and do not establish a general ranking of loss functions; boldface marks the highest score within each detector family.

Across the official test results, the effect depends on the detector and metric. In the canonical TinyPerson comparison, FCOS with the full H-WIoU bundle raises tiny1 AP₅₀ from 9.95% to 15.71%, while Cascade R-CNN has nearly unchanged tiny1 AP₅₀ and lower overall AP₅₀ and AP₇₅. On AI-TOD-v2, FCOS H-WIoU raises AP from 16.65% to 17.16%, while AP for very tiny objects falls from 4.05% to 3.26%. These results describe the tested Seed-42 checkpoints and do not show a uniform benefit across models.

Figure 3 summarizes the metric spread across all 71 canonical and extended test evaluations; exact values are listed in Tables 6, 7, 8, and 9. Each marker represents one checkpoint; color identifies detector family, while circles and triangles distinguish the 20 canonical from the 51 additional models. The left panel plots AI-TOD-v2 AP against AP₅₀ for 32 models; the right plots TinyPerson overall AP₅₀ against tiny1 AP₅₀ for 39 models. These benchmark-specific metric plots summarize tested checkpoints and do not establish a statistical ranking.

5.9 Computational Efficiency and Inference Latency

Because the proposed loss and assignment terms are used only during training, they introduce no additional inference-time operators. At test time, the network architecture, number of parameters, and FLOPs are identical to the corresponding baselines.

Table 8: Additional AI-TOD-v2 official-test results.

Detector	Formulation	AP	AP ₅₀	AP ₇₅	AP _{vt}	AP _t	AP _s	AR ₁₅₀₀
FCOS	Alpha-IoU	5.89	16.80	3.13	0.24	3.58	10.24	13.26
FCOS	DIoU	5.96	17.28	2.85	0.31	3.99	10.46	13.74
FCOS	EIoU	6.09	17.34	2.98	0.30	3.71	10.67	13.92
FCOS	GCD	5.98	16.80	3.26	0.31	3.73	10.34	13.48
FCOS	GIoU	6.17	17.19	3.11	0.27	3.87	10.71	13.68
FCOS	IGWD	5.69	16.76	2.74	0.23	3.72	9.74	13.22
FCOS	Inner-IoU	6.01	17.11	3.03	0.27	3.87	10.37	13.81
FCOS	NWD	5.69	16.23	2.83	0.29	3.68	10.12	13.20
FCOS	SimD	5.74	16.70	2.93	0.28	3.54	9.90	12.89
FCOS	SIoU	5.96	17.11	2.97	0.29	3.83	10.32	13.76
FCOS	WIoU	6.27	17.57	3.27	0.29	3.82	10.91	13.76
Faster R-CNN	ATSS	8.72	22.12	4.95	2.64	9.41	11.67	17.87
Faster R-CNN	HLA, RPN only	21.27	48.58	15.03	8.90	21.32	24.64	31.33
Faster R-CNN	NWD assignment	18.43	42.00	12.95	8.85	19.27	19.85	27.68
Faster R-CNN	RFLA	18.21	41.34	12.74	7.91	18.43	20.69	26.83
Faster R-CNN	SimD assignment	20.72	45.59	15.31	8.58	20.09	24.53	29.39
YOLO26s	H-WIoU	20.39	48.81	13.46	6.25	18.63	27.18	29.97
YOLO26s	IGWD	20.95	48.51	14.67	7.22	19.66	27.35	30.83
YOLOv8s	CIoU	21.72	49.86	15.21	7.08	19.87	28.39	32.00
YOLOv8s	H-TAL	20.80	49.86	13.68	6.21	18.68	28.66	31.94
YOLOv8s	H-WIoU	21.53	50.52	14.48	6.66	19.45	28.36	31.19
YOLOv8s	IGWD	21.82	50.27	15.43	6.71	20.17	28.27	32.10

Table 10 reports forward-pass latency and throughput on an NVIDIA RTX 5070 Ti, measured over all 1,684 real Scale Match TinyPerson validation tiles. Each model uses batch size 1, with 50 warmup inputs followed by 1,684 timed inputs; tiles are bilinearly resized to the listed input resolution. Timing covers model forward passes and excludes image loading, resizing, and host-to-device transfer. Latency differences are relative to the family’s baseline; boldface marks the highest FPS and lowest latency within each family.

Because H-WIoU, HLA, H-TAL, and H-STAL are applied during training, they do not introduce additional inference-time modules. On the tested RTX 5070 Ti, measured forward-time differences on TinyPerson validation tiles range from -4.7% to +1.2% relative to the corresponding baselines. These measurements are consistent with the unchanged inference computation, but should not be interpreted as evidence that the proposed method makes inference faster.

6 Ablations and Factorial Analysis

6.1 Faster R-CNN Placement Breakdown

Table 11 breaks down the effect of applying scale homotopy to RPN assignment versus RoI regression in Faster R-CNN. Boldface marks the highest value in each metric column.

As shown in Table 11 and Figure 4, modifying RPN assignment alone (HLA) yields the highest overall AP₅₀ (42.48%). Meanwhile, combining HLA with RoI H-WIoU (Dual placement) produces the best microscopic AP₅₀^{tiny1} (25.65%) and reasonable-scale AP₅₀ (61.75%).

Table 9: Additional TinyPerson official-test results.

Detector	Formulation	AP ₂₅ ^{all}	AP ₅₀ ^{all}	AP ₇₅ ^{all}	AP ₅₀ ^{tiny}	AP ₅₀ ^{tiny1}	AP ₅₀ ^{tiny2}	AP ₅₀ ^{tiny3}	AP ₅₀ ^{small}	AP ₅₀ ^{reas.}
FCOS	GIoU baseline	65.44	41.55	6.13	34.44	11.55	34.71	49.58	61.11	61.43
FCOS	EH-WIoU	69.66	43.75	6.36	38.07	15.79	39.52	52.65	61.49	60.74
FCOS	Alpha-IoU	64.95	40.80	5.94	34.16	12.18	34.00	49.43	59.89	59.56
FCOS	CIoU	65.07	40.78	5.80	34.42	11.21	33.61	50.67	60.96	60.22
FCOS	DIoU	65.46	40.38	5.28	34.12	13.39	34.91	48.38	58.02	58.94
FCOS	EIoU	63.09	38.34	5.00	30.67	9.72	29.68	46.67	58.56	60.79
FCOS	FCOS factorial (GIoU with TOD sampling)	63.24	40.25	5.59	32.48	11.31	30.24	49.04	61.28	61.50
FCOS	H-WIoU (standard centerness and sampling)	63.91	39.95	5.80	32.22	9.20	31.67	48.12	61.15	61.15
FCOS	H-WIoU + TOD sampling	62.52	39.29	5.51	31.40	10.83	30.44	47.03	60.23	61.07
FCOS	H-WIoU ($\sigma_0 = 4$)	63.75	40.08	5.84	31.69	9.61	30.72	47.47	61.04	62.84
FCOS	H-WIoU ($\sigma_0 = 8$)	64.14	39.76	5.79	32.38	10.06	31.08	48.41	60.95	60.01
FCOS	GCD	62.69	37.70	5.31	32.24	9.97	32.55	46.94	56.20	53.67
FCOS	IGWD	63.15	38.87	5.14	31.20	8.89	29.96	47.39	59.65	60.95
FCOS	Inner-IoU	63.91	39.93	5.66	32.08	9.63	31.06	48.25	60.69	62.18
FCOS	NWD	61.29	35.50	4.44	30.46	10.90	29.76	44.10	53.11	50.27
FCOS	SimD	63.98	40.53	5.51	32.91	9.72	31.30	49.24	61.84	61.66
FCOS	SIoU	62.63	39.02	5.46	30.62	9.44	29.94	45.86	59.92	61.54
FCOS	WIoU	64.98	40.73	5.82	34.65	11.90	35.53	49.73	60.12	59.79
Faster R-CNN	ATSS	68.68	50.23	9.47	45.18	23.49	48.73	59.43	68.31	65.70
Faster R-CNN	NWD	69.28	51.26	10.01	45.42	23.31	48.04	60.02	69.09	69.48
Faster R-CNN	RFLA	71.07	50.93	9.61	46.98	25.91	49.58	61.11	67.86	65.43
Faster R-CNN	SimD	71.37	52.88	10.49	48.50	28.88	51.67	61.37	69.37	68.64
YOLO26s	H-STAL (NMS-free)	72.94	52.41	8.35	48.90	34.97	55.72	60.15	64.83	61.37
YOLO26s	H-WIoU	71.42	51.46	8.84	49.04	33.93	53.24	59.68	64.86	57.85
YOLO26s	IGWD	71.91	52.76	9.82	51.72	38.91	56.48	60.10	63.24	58.93
YOLOv8s	CIoU	72.37	53.84	9.85	51.91	36.57	57.79	61.35	65.36	60.31
YOLOv8s	H-TAL	71.87	51.68	7.81	49.04	33.88	55.30	61.35	64.46	59.18
YOLOv8s	H-WIoU	72.12	53.00	9.37	50.56	33.90	55.74	61.74	66.29	60.41
YOLOv8s	IGWD	72.19	53.51	10.06	51.49	35.47	57.23	61.68	65.20	59.65

Table 10: TinyPerson inference efficiency.

Family	Variant	Input Res.	Params	GFLOPs	Latency (ms)	FPS	Latency Δ
Faster R-CNN	Baseline	512×512	41.3M	70.18	18.41 ± 2.42	54.3	—
Faster R-CNN	EH-WIoU Dual	512×512	41.3M	70.18	17.55 ± 0.80	57.0	-4.7%
FCOS	GIoU Baseline	512×512	32.1M	51.34	14.89 ± 0.88	67.2	—
FCOS	H-WIoU + Gaussian centerness	512×512	32.1M	51.34	14.85 ± 0.94	67.4	-0.3%
FCOS	H-WIoU + Gaussian centerness + TOD sampling	512×512	32.1M	51.34	15.05 ± 1.00	66.5	+1.1%
Cascade R-CNN	Baseline	512×512	41.9M	118.35	20.03 ± 0.84	49.9	—
Cascade R-CNN	EH-WIoU	512×512	41.9M	118.35	20.27 ± 0.81	49.3	+1.2%
RetinaNet	Baseline	512×512	32.2M	52.17	11.00 ± 0.93	90.9	—
RetinaNet	H-WIoU	512×512	32.2M	52.17	11.01 ± 1.02	90.9	+0.05%
YOLOv8s	Baseline	640×640	11.1M	14.33	4.86 ± 0.63	205.7	—
YOLOv8s	H-TAL	640×640	11.1M	14.33	4.87 ± 0.66	205.3	+0.2%
YOLO26s	Baseline	640×640	9.9M	10.39	7.53 ± 0.79	132.7	—
YOLO26s	H-STAL (NMS-Free)	640×640	9.9M	10.39	7.61 ± 0.75	131.4	+1.0%

6.2 FCOS Factorial Analysis

We analyze three FCOS factors: the regression loss (GIoU or H-WIoU), centerness (standard or Gaussian), and sampling (standard or TOD-adaptive). Table 12 displays all eight combinations from the frozen Seed-42 protocol. Each row uses its best validation AP₅₀ checkpoint and passed full-validation replay; boldface marks the highest score in each metric column.

Figure 5 plots overall and tiny1 AP₅₀ for the eight arms from the hash-bound factorial manifest and accepted full-validation replay.

Stabilizing Degenerate Training. Under the evaluated FCOS protocol, the two GIoU configurations with Gaussian centerness produce zero AP. The corresponding H-WIoU configurations do not: H-WIoU with Gaussian centerness reaches 31.13% overall AP₅₀, while adding TOD-adaptive sampling reaches 11.92% tiny1 AP₅₀. This comparison shows an interaction between the regression

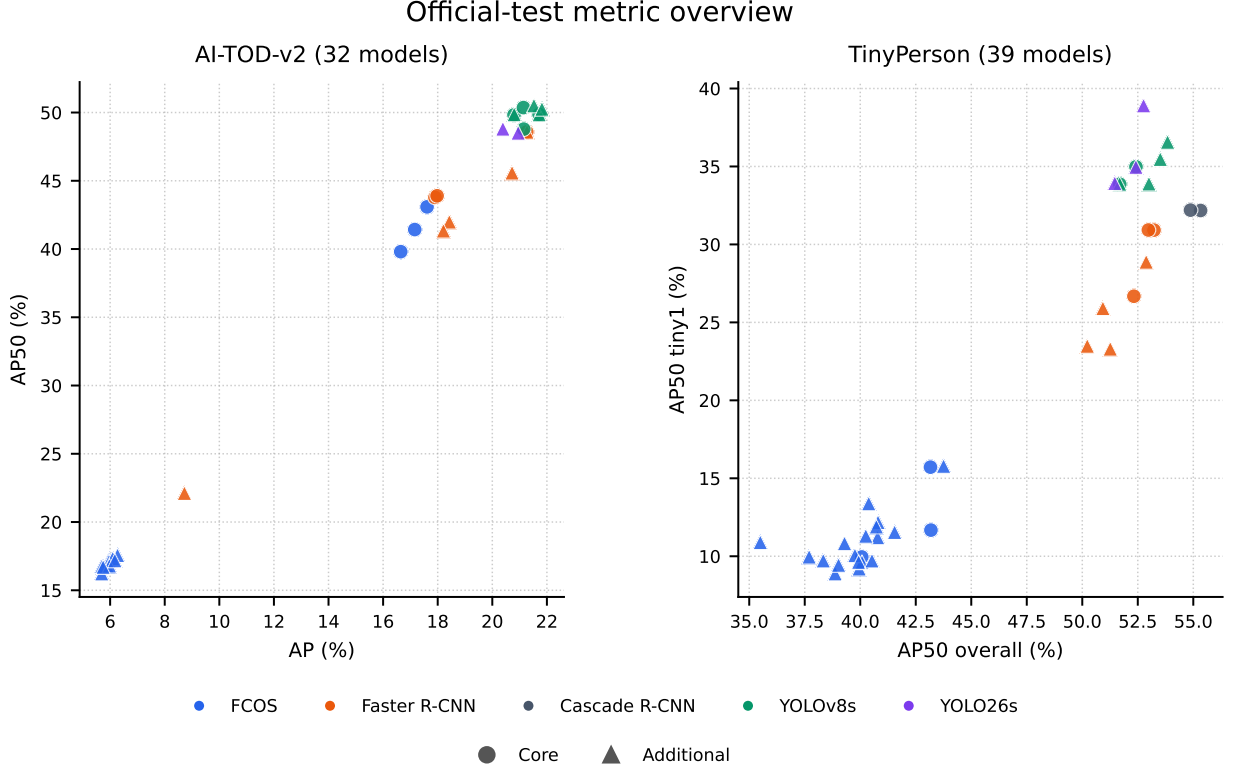


Figure 3: Metric overview of canonical and additional official-test models.

Table 11: Faster R-CNN placement ablation on TinyPerson.

Placement Strategy	AP_{25}^{all}	AP_{50}^{all}	AP_{75}^{all}	AP_{50}^{tiny1}	$AP_{50}^{\text{reas.}}$
Baseline	59.30	42.22	5.92	20.35	54.59
RoI Loss Only	58.37	41.12	5.55	23.72	56.93
RPN HLA Only	58.57	42.48	6.12	22.59	54.52
Dual (HLA + RoI H-WIoU)	59.48	42.22	6.05	25.65	61.75

objective and Gaussian centerness; it does not isolate the mechanism behind the zero-AP results. Figure 5 illustrates this pattern.

6.3 Entropy Coupling Factor (β)

Sweeping the entropy coupling parameter $\beta \in \{0, 0.25, 0.5, 0.75, 1.0\}$ in Faster R-CNN yields overall AP_{50} values of 42.23%, 42.39%, 42.48%, 42.47%, and 42.97%, with corresponding tiny1 values of 26.32%, 24.92%, 22.59%, 24.52%, and 26.46%. Setting $\beta = 1.0$ improves overall AP_{50} by +0.74 pp over the uncoupled $\beta = 0$ reference, while reasonable-scale AP increases from 56.62% to 59.78% (+3.16 pp).

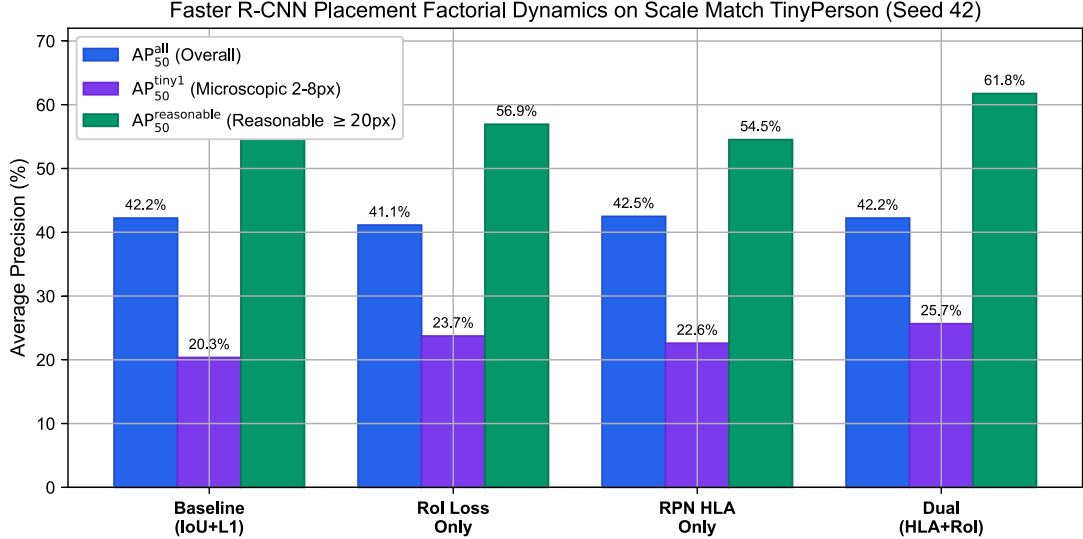


Figure 4: Faster R-CNN placement results on TinyPerson.

Table 12: FCOS factorial results on TinyPerson.

Configuration	AP_{50}^{all}	AP_{50}^{tiny1}	$AP_{50}^{reas.}$
GIoU baseline	27.52	5.58	43.45
GIoU + TOD sampling	27.89	6.94	46.60
GIoU + Gaussian centerness	0.00	0.00	0.00
GIoU + Gaussian centerness + TOD sampling	0.00	0.00	0.00
H-WIoU (standard settings)	26.23	5.66	40.48
H-WIoU + TOD sampling	27.23	6.34	44.24
H-WIoU + Gaussian centerness	31.13	8.10	51.26
H-WIoU + Gaussian centerness + TOD sampling	30.10	11.92	41.47

7 Discussion

Across the reported validation and official-test comparisons, some H-WIoU configurations improve tiny-object scores, while changes in aggregate scores depend on the detector and training setup. On the official test splits, TinyPerson FCOS with the full bundle has higher tiny1 AP50 than its baseline, but the Cascade R-CNN comparison has slightly lower overall AP50. On AI-TOD-v2, the FCOS H-WIoU model improves overall AP over its baseline but has a lower very-tiny AP. These single-seed results describe the tested checkpoints; they do not establish a universal advantage or explain why a difference occurs.

8 Limitations and Boundaries

H-WIoU has several important boundaries. First, the centroid-gradient lower bound applies only to the uncapped SNGD and target-normalized W_2 affinities on compact, strictly disjoint box domains in exact real arithmetic; it is not a global, scale-uniform guarantee and does not imply convergence of the full detector. It does not cover YOLO’s capped exponent or numerical saturation. Second, all reported experiments use a single random seed (42), so run-to-run variance and statistical

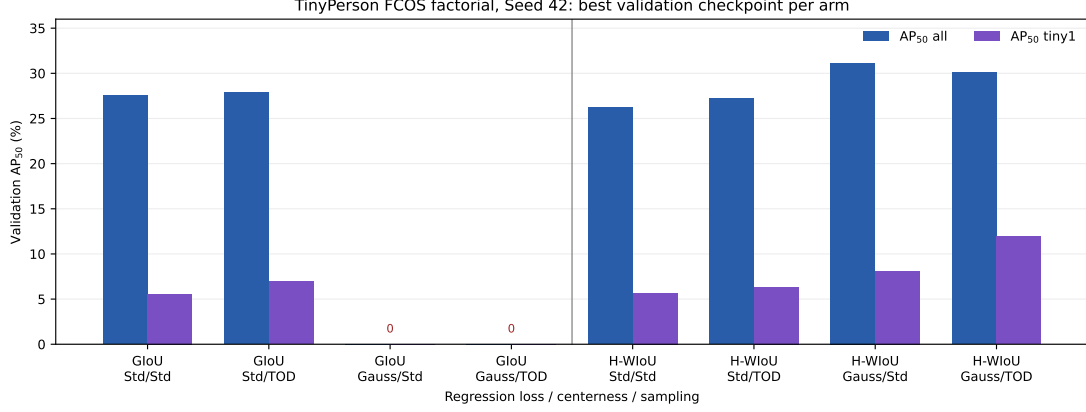


Figure 5: FCOS factorial AP₅₀ by configuration.

significance are not estimated. Third, SNGD uses logarithms while the target-normalized W_2 route does not; both use an exponential affinity and add training-time computation, which is not separately benchmarked here. Finally, the empirical effect is detector- and configuration-dependent: several experiments show improvements on tiny-object subsets together with neutral or negative changes in aggregate metrics.

9 Conclusion

We studied Scale-Adaptive Wasserstein-IoU (H-WIoU) as a family that blends an arm-specific scale-normalized geometric affinity with IoU using a target-scale-dependent gate. The gradient result is limited to the uncapped SNGD and target-normalized W_2 variants on the compact, strictly disjoint box domain stated in the analysis. We applied the identified similarities to regression and label assignment in several detector families.

Across 71 official-test evaluations from TinyPerson and AI-TOD-v2, the observed effects vary by detector and metric. Several tested configurations score higher on selected tiny-object metrics, while other aggregate scores are unchanged or lower. In the single-seed TinyPerson homotopy-form comparison, the sigmoid form had higher validation AP, AP₅₀, and tiny1 AP₅₀ than the rational form. The results support further study of scale-conditioned box similarity, but do not establish that the rational gate is the best choice or that H-WIoU improves every detector. Since all experiments use Seed 42, run-to-run variation is not measured. The proposed terms are used during training and add no inference-time operators.

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A Qualitative Detections

Qualitative examples from the Scale Match TinyPerson validation split are presented in Figure 6. The historical AI-TOD-v2 sigma-4 examples are excluded because that run has an unresolved frozen-versus-runtime protocol identity conflict.

Predictions use a score threshold of 0.30 and class-aware NMS at IoU 0.50. Green dashed boxes mark ground-truth targets; orange and blue boxes show matched baseline and proposed predictions; red boxes mark unmatched false positives. The proposed configuration combines H-WIoU with Gaussian centerness and TOD-adaptive sampling, so these examples do not isolate the loss effect. In the microscopic crowded scene, correct matches rise from 3 to 7 with false positives rising from 13 to 38. In the dense seaside scene, detections rise from 8/36 to 17/36 with false positives rising from 21 to 45. In the limitation case, both configurations detect 1/2 targets, with 8 and 7 false positives, respectively.

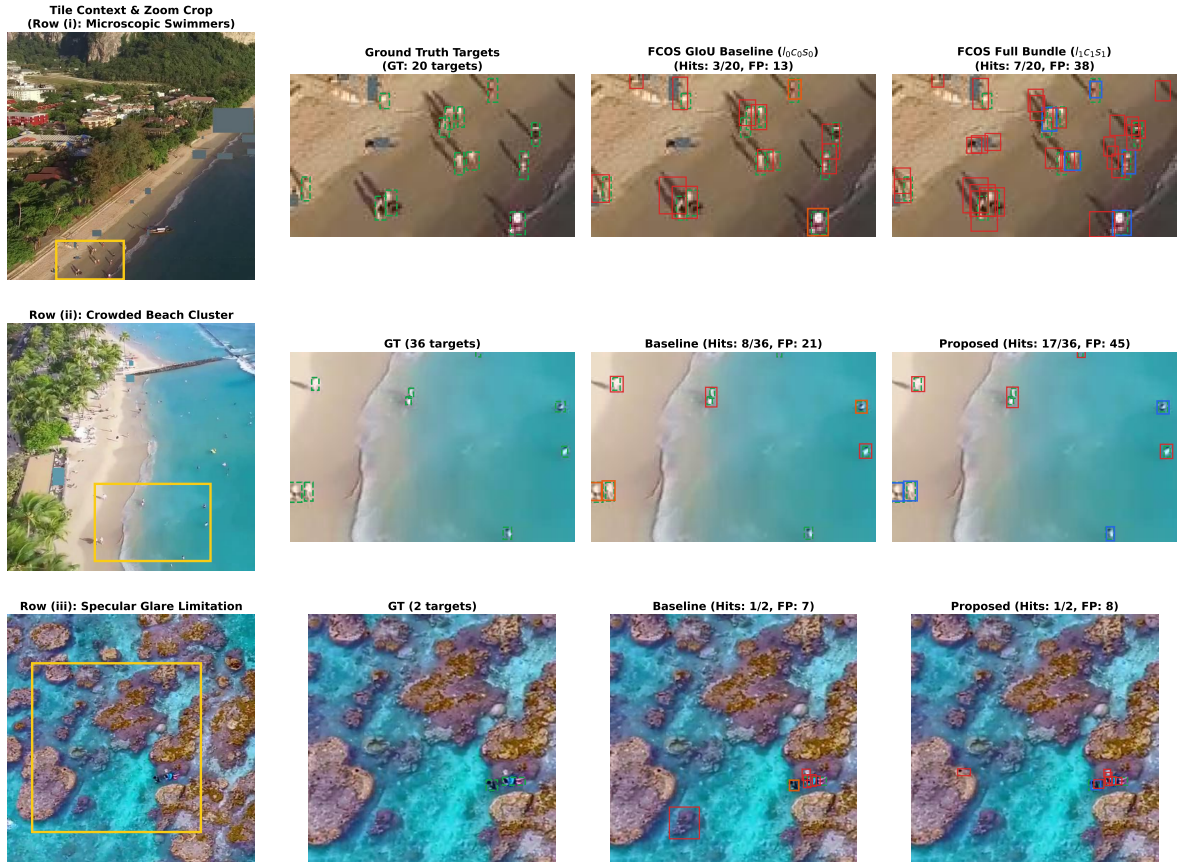


Figure 6: Qualitative TinyPerson validation examples.